INCH-POUND

A-A-58060C
28 Jul 2021

SUPERSEDING
A-A-58060B
14 Jan 2011

COMMERCIAL ITEM DESCRIPTION

FLUOROCARBON AND OTHER REFRIGERANTS

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description covers the requirements for fluorocarbon and other refrigerants. Most of the refrigerants listed have ceased new production by 2010 since they are known to deplete ozone. Refrigerants may still be acquired from the re-capture and/or recycling of chlorofluorocarbons (CFCs) (see 4.1).

2. CLASSIFICATION. The refrigerants shall conform to the American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) designated numbers (R number) as listed in Table I.

TABLE I. Classification of refrigerants.

R number	Chemical Name (mass % composition; 100% if blank)
R11	Trichloromonofluoromethane
R12	Dichlorodifluoromethane
R13	Monochlorotrifluoromethane
R22*	Monochlorodifluoromethane
R23	Trifluoromethane
R32	Difluoromethane
R113	Trichlorotrifluoroethane
R114	Dichlorotetrafluoroethane
R116	Hexafluoroethane
R123	Dichlorotrifluoroethane
R124	Monochlorotetrafluoroethane
R125*	Pentafluoroethane

Comments, suggestions, or questions on this document should be addressed to AFPET/PTPS, 2430 C Street, Building 70, Area B, Wright-Patterson AFB OH 45433-7631 or e-mailed to.afpet.ptps@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6830

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:59Z
Check the source to verify that this is the current version before use.

TABLE I. Classification of refrigerants - Continued.

R number	Chemical Name (mass % composition, 100% if blank)
R134a*	Tetrafluoroethane
R143a	Trifluoroethane
R236fa	Hexafluoropropane
R401A	Monochlorodifluoromethane (53%); Difluoroethane (13%); Monochlorotetrafluoroethane (34%)
R401B	Monochlorodifluoromethane (61%); Difluoroethane (11%); Monochlorotetrafluoroethane (28%)
R402A	Pentafluoroethane (60%); Propane (2%); Monochlorodifluoromethane (38%)
R402B	Pentafluoroethane (38%); Propane (2%); Monochlorodifluoromethane (60%)
R404A*	Pentafluoroethane (44%); Trifluoroethane (52%); Tetrafluoroethane (4%)
R405A	Monochlorodifluoroethane (45%); Difluoroethane (7%); Monochlorodifluoroethane (5.5%); Octafluorocyclobutane (42.5%)
R406B	Monochlorodifluoromethane (55%); Isobutane (4%); Monochlorodifluoroethane (41%)
R407A*	Difluoromethane (20%); Pentafluoroethane (40%); Tetrafluoroethane (40%)
R407B*	Difluoromethane (10%); Pentafluoroethane (70%); Tetrafluoroethane (20%)
R407C*	Difluoromethane (23%); Pentafluoroethane (25%); Tetrafluoroethane (52%)
R408A	Pentafluoroethane (7%); Trifluoroethane (46%); Monochlorodifluoromethane (47%)
R409A	Monochlorodifluoromethane (60%); Monochlorotetrafluoroethane (25%); Monochlorodifluoroethane (15%)
R410A*	Difluoromethane (50%); Pentafluoroethane (50%)
R410B*	Difluoromethane (45%); Pentafluoroethane (55%)
R411A	Propylene (1.5%); Monochlorodifluoromethane (87.5%); Difluoroethane (11%)
R411B	Propylene (3%); Monochlorodifluoromethane (94%); Difluoroethane (3%)
R412A	Monochlorodifluoromethane (70%); Octafluoropropane (5%); Monochlorodifluoroethane (25%)
R416A	Tetrafluoroethane (59%); Monochlorotetrafluoroethane (39.5%); N-Butane (1.5%)
R500	Dichlorodifluoromethane (74%); Difluoroethane (26%)
R502	Monochlorodifluoromethane (49%); Monochloropentafluoroethane (51%)
R503	Trifluoromethane (40%); Monochlorotrifluoromethane (60%)
R507A*	Pentafluoroethane (50%); Trifluoroethane (50%)
R508A	Trifluoromethane (39%); Hexafluoroethane (61%)
R509A	Monochlorodifluoromethane (44%); Octafluoropropane (56%)

*May not be available under EPA Final Rule 2016 "Protection of Stratospheric Ozone: New Listings of Substitutes; Changes of Listing Status; and Reinterpretation of Unacceptability for Closed Cell Foam Products Under the Significant New Alternatives Policy Program," and Revision of Clean Air Act Section 608 "Venting Prohibition for Propane," *Federal Register, Vol. 81, No. 241*.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:59Z
Check the source to verify that this is the current version before use.

3. SALIENT CHARACTERISTICS. Refrigerants will conform to Air-Conditioning, Heating, and Refrigeration Institute (AHRI) Standard 700, Specifications for Fluorocarbon Refrigerants, when tested according to Appendix C of this same standard.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The manufacturer/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR). Note that recovered materials will require purification to meet the requirements of AHRI Standard 700.

4.2 Safety data sheet (SDS). Manufacturers shall prepare and submit a SDS in accordance with FED-STD-313, "Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities" and meeting the requirements of the Code of Federal Regulations (CFR), 29 CFR 1910.1200, "Hazard Communication."

4.3 Hazard Warning Label. Occupational Safety and Health Administration hazard warning labels shall be furnished in accordance with 29 CFR 1910.1200.

4.4 Transportation. Manufacturers and users of compressed gas cylinders shall comply with all applicable Department of Transportation (DOT) regulations and MIL-STD-1411 "Inspection and Maintenance of Compressed Gas Cylinders" for the inspection, test, storage, transportation, environmental impact, ownership accountability, and disposal of compressed gas cylinders and their contents.

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market Acceptability. The product offered must have been previously sold either to the Government or in the commercial market.

6. PACKAGING.

6.1 Compressed gas cylinders. The specific use of any given cylinder for the packaging of products may dictate special cylinder design and construction requirements. For this reason, all cylinders used for the packaging of products shall conform to all special restrictions to cylinder design and manufacturing techniques imposed by the procuring activity.

6.1.1 Reusable cylinders and valves. Unless otherwise specified herein, the refrigerants shall be contained in Government-furnished cylinders conforming to 49 CFR 100-199. When specified (see section 7), cylinders and valves shall be furnished by the contractor and, for military users, shall conform to A-A-59666 "Cylinder, Compressed Gas: DOT Specifications 4B, 4BA, 4BW, and 4E" and A-A-59860 "Valves, Cylinder, Gas (for Compressed or Liquefied Gases)," respectively. When specified (see section 7), the refrigerants shall be contained in contractor-owned, DOT-approved containers with valves and with Compressed Gas Association (CGA) 660 outlet connections.

6.1.2 Disposable cylinders. When specified (see section 7), the contractor shall furnish disposable containers of the capacity as specified conforming to 49 CFR 178.65 "Specification 39 non-reusable (non-refillable) cylinders."

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:59Z
Check the source to verify that this is the current version before use.

6.1.3 Cylinder maintenance. The contractor shall inspect, maintain, and fill Government furnished cylinders according to MIL-STD-1411.

6.2 Drums (types R11, R113, and R123). These refrigerants shall be shipped in DOT approved containers. When specified (see section 7), drums conforming to UN type containers according to Title 49, Part 178, Subpart L, shall be used.

7. NOTES.

7.1 Intended use. Refrigerants covered by this specification are primarily intended for use in refrigeration and air conditioning units; however, some refrigerants may be used as nonpolar heat transfer, fire extinguishing, and electronic isolation media.

7.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. CID document number, revision, and CID PIN (see section 7.3).

b. Type of refrigerant required (section 2).

c. When the contractor shall furnish Government-approved cylinders (section 6.1.1).

d. When the contractor shall furnish DOT-approved containers (section 6.1.2).

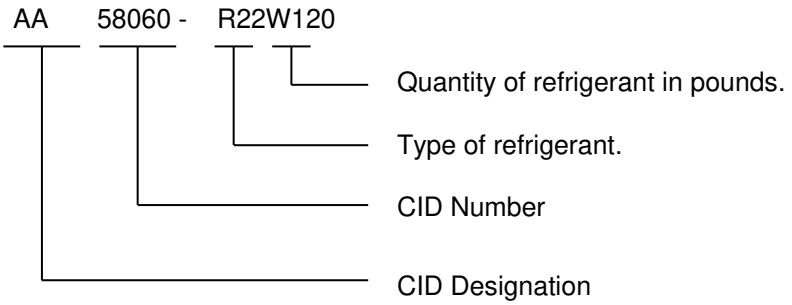
e. When the contractor shall provide contractor-owned cylinders (section 6.1.1).

f. When cylinders conforming to DOT specification 39 are required (section 6.1.2).

g. When drums are required (section 6.2).

7.3 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.

This example describes a part number system for CID A-A-58060, 120 pounds of R-22 refrigerant.



7.4 Source of documents.

7.4.1 Air-Conditioning, Heating, and Refrigeration Institute. Copies of AHRI Standard 700 may be obtained from https://www.ahrinet.org.

7.4.2 Federal Acquisition Regulations. Electronic copies of FAR documents may be obtained from https://www.acquisition.gov/browse/index/far.

7.4.3 Federal regulations. Electronic copies of CFR documents may be obtained from https://ecfr.gov/.

7.4.4 Military specifications and standards. Electronic copies of federal specifications, standards, and CIDs are available online at https://quicksearch.dla.mil.

7.4.5 Compressed Gas Association. Electronic copies of CGA publications may be obtained from https://www.cganet.com/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:59Z
Check the source to verify that this is the current version before use.

7.4.6 ASHRAE Refrigerant Designations. A list of refrigerant designations may be found at https://ashrae.org/technical-resources/standards-and-guidelines/ashrae-refrigerant-designations.

7.5 Key Words.

Cylinders
Difluoroethane
Difluoromethane
Drums
Fire extinguishing
Monochlorodifluoromethane
Monochlorotetrafluoroethane
Pentafluoroethane
Propane
Propylene
Refrigerant

7.6 Changes from previous issues. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

MILITARY INTERESTS:

Custodian:
Army – AV
Navy – SH
Air Force – 68
DLA – GS

Preparing Activity:
Air Force – 68
(Project 6830-2021-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:59Z
Check the source to verify that this is the current version before use.